SWV Forward Model

From Cleavage Fraction to Recorded Voltammogram

Eton iGEM 2026, Dry Lab | Synthesis of Layers 6, 8, 9, 10, 11, 13 | `swv_forward_model.py`

I. What This Model Does

The physical story

Picture the electrode surface. Thousands of peptide chains are chemisorbed to a gold disc through a short alkyl spacer. Each chain ends in a methylene blue (MB) dye molecule. MB is a redox reporter: it can gain or lose electrons, and we detect that electrochemically. MMP-9, if present in the wound fluid, cuts each peptide at a specific bond. The MB label falls away. The electrochemical signal drops.

The SWV forward model answers one question precisely: **if a fraction F_k of those peptide chains have been cleaved, what voltammogram would we actually record?**

Answering this requires three pieces of physics, linked in sequence:

1. **Tether geometry.** MB is not at a fixed distance from the gold surface. The peptide chain is flexible and wiggles thermally, so MB samples a range of distances. We need the probability distribution $P(d)$ of those distances, because the electron transfer rate depends exponentially on d.
2. **Electron transfer kinetics.** Once we know d, we can calculate how fast the redox reaction is. This rate $k_0(d)$ sets how MB responds to the SWV voltage waveform.
3. **BV-SWV simulator.** Given the rate k_0, a Butler–Volmer model converts the SWV waveform into the net current $\Delta I(E)$ that the potentiostat records. Averaging over the $P(d)$ distribution and scaling by $(1 - F_k)$ gives the complete voltammogram.

The chain is therefore:

$$[\text{MMP-9}],\ t \xrightarrow{\text{kinetics (Sec. VII)}} F_k \xrightarrow{\text{tether + ET (Sec. II–III)}} k_0(d),\ P(d) \xrightarrow{\text{BV-SWV (Sec. IV–V)}} \Delta I(E)$$

How to read this report

Each section opens with the physical picture, then derives the equations that formalise it, and closes with a **Key result** box. Every equation is followed by a plain-English gloss beginning with *This says.* Parameters are tabulated at the section where they first appear. The code implementing all steps is in `swv_forward_model.py`.

All parameter values are confirmed from papers physically present in `papers/swv_forward_model/` or cross-referenced from other layer paper folders as noted. Values that remain unconfirmed are labelled **[FLAG: unconfirmed]**.

II. Tether Geometry: Where Does MB Sit?

Physical picture

The gold electrode surface is coated with a mixed self-assembled monolayer (SAM). The active component is our peptide probe: a PEG6 spacer (six ethylene oxide units) anchors to the gold via a thiol group, the ten-residue peptide SGKIPRTLTA extends from the PEG6 tip, and MB is attached at the far end. When only the peptide probe is present, 6-mercaptohexanol (MCH) passivates the background surface; when the Fc reference is co-deposited, MCH is omitted (Lab note, Jaden Potishko, Apr 2026).

At body temperature in wound fluid, the peptide chain is flexible. Rather than sitting at a fixed distance, MB thermally fluctuates across a range of distances from the gold surface. Figure 1C shows this: $P(d)$ is the probability density of finding MB at distance d from the gold.

Why does this matter? Because electron transfer probability falls exponentially with distance. A probe with MB 10 Å from the gold has a rate orders of magnitude higher than one with MB at 25 Å. The voltammogram is the average over all these probes, and that average is dominated by the shape of $P(d)$.

Freely jointed chain (FJC) model

We model the peptide as a **freely jointed chain**: N_K rigid segments of length b joined at random orientations, like links in a necklace. This is the simplest polymer model that captures the key physics of an unstructured flexible chain. More elaborate models (worm-like chain, all-atom MD) are listed in Section XI as future work.

Parameter	Symbol	Value	Source
Residue count	N_{res}	10	SGKIPRTLTA
Contour length per residue	ℓ_0	0.38 nm	standard
Kuhn length	b	0.76 nm	Kohn 2004 [2]
PEG6 spacer (Flory, dilute)	d_{PEG}	1.40 nm	Layer 8 model

Since each Kuhn segment spans two residues ($b = 2\ell_0$), the number of Kuhn segments in the ten-residue peptide is $N_K = N_{\text{res}}/2 = 5$.

The mean-squared end-to-end distance of the chain:

$$\langle r^2 \rangle = N_K\, b^2 = 5 \times (0.76)^2 = 2.888 \text{ nm}^2 \tag{II.1}$$

▷ *a longer or stiffer chain (larger $N_K b^2$) pushes MB farther from the surface on average.*

Because the chain is isotropic, the three spatial directions (x, y, z) are equivalent. The standard deviation of MB distance along the surface-normal axis (z, pointing away from gold) is:

$$\sigma_z = \sqrt{\langle r^2 \rangle / 3} = \sqrt{2.888/3} = 0.981 \text{ nm} \tag{II.2}$$

▷ *σ_z is the typical spread of MB positions above the PEG6 tip. It is roughly 1 nm, set by the peptide chain entropy.*

The distance distribution $P(d)$

The PEG6 spacer holds the base of the peptide at a mean height $d_{\text{PEG}} = 1.40$ nm above the gold surface (Flory radius in dilute regime, Layer 8 model at $\Gamma_p = 3$ pmol cm^{-2}). This acts as an effective hard wall for the peptide FJC. Above this, the peptide chain end-position follows a 3D Gaussian shifted by d_{PEG}, giving a half-Gaussian:

$$P(d) = \frac{\sqrt{2/\pi}}{\sigma_z} \exp\left(-\frac{(d - d_{\text{PEG}})^2}{2\sigma_z^2} \right), \qquad d \geq d_{\text{PEG}} \tag{II.3}$$

▷ *the most likely position for MB is just above the PEG6 tip ($d_{PEG} = 14$ Å); it becomes progressively less likely farther away, with spread $\sigma_z \approx 10$ Å.*

The mean distance (centre of mass of the distribution):

$$\langle d \rangle = d_{\text{PEG}} + \sigma_z \sqrt{2/\pi} = 1.398 + 0.784 = 2.182 \text{ nm} = 21.8 \text{ Å} \tag{II.4}$$

III. Electron Transfer Kinetics: How Fast Does MB React?

Physical picture

For MB to undergo redox chemistry, an electron must tunnel through the molecular bridge between the electrode and MB. Tunnelling probability falls off exponentially with distance: every extra Ångström makes tunnelling roughly $e^{-1} = 37\%$ as likely. This means the electron transfer rate k_0 is extremely sensitive to exactly where MB sits, which is why the $P(d)$ distribution calculated above matters so much.

The tunnelling rate also depends on the medium: electrons tunnel faster through conjugated aromatic systems than through saturated alkyl chains. The PEG6 spacer plus peptide backbone is an insulating bridge, so the decay is rapid. The exact β for PEG-linked probes has not been confirmed from our paper set (Smalley 1995 [3] characterises alkanethiol SAMs, not PEG). This is flagged as the key experimental parameter to measure at Layer 15 via EIS.

Tunnelling decay equation

Parameter	Symbol	Value	Source
Tunnelling decay constant	β	1.0 Å^{-1}	Smalley 1995 [3] **[FLAG: unconfirmed]**
Rate at PEG6 tip	k_{ref}	≈ 127 s^{-1}	derived **[FLAG: unconfirmed]**

The rate at any distance d beyond the PEG6 tip:

$$k_0(d) = k_{\text{ref}} \exp\big(-\beta(d - d_{\text{PEG}})\big) \tag{III.1}$$

▷ *each additional Ångström of distance beyond the PEG6 tip reduces the rate by a factor* $e^{-1.0} = 0.37$. *The FLAG on* β *means this is an estimate until EIS is done.*

The reference rate at the PEG6 tip, using $\beta = 1.0$ Å^{-1} (alkanethiol value as placeholder):

$$k_{\text{ref}} = 1.5 \times 10^8 \times \exp\big(-1.0 \times 14.0\big) = 127 \text{ s}^{-1} \tag{III.2}$$

Implication for κ: At d_{PEG} (PEG6 tip), $k_0 = 127$ s^{-1} and $\kappa = 4.2$ at 15 Hz (quasi-reversible, signal exists). At the mean peptide distance $\langle d \rangle = 21.8$ Å, $k_0 = 0.05$ s^{-1} and $\kappa = 0.002$ (deep irreversible, negligible contribution). The SWV signal therefore comes predominantly from probes near d_{PEG} (the P(d) peak), not from the mean-distance probes. This is the key difference from the C6 alkyl case: the PEG6 spacer shifts the dominant signal population to the $P(d)$ peak rather than the mean.

The dimensionless kinetic parameter κ

Whether MB responds measurably to the SWV waveform depends not on k_0 alone, but on how fast it is relative to the SWV pulse duration. This comparison is captured by the dimensionless parameter:

$$\kappa = k_0 \times \tau = \frac{k_0}{2f} \tag{III.3}$$

▷ κ *is the number of redox events that can happen during one SWV half-pulse of duration* $\tau = 1/(2f)$.

There are three regimes:

- $\kappa \gg 1$ (reversible): the reaction equilibrates fully during the pulse. Net current $\Delta I \to 0$ because no charge accumulates between pulses. Sensitivity is low.
- $\kappa \ll 1$ (irreversible): the reaction barely starts during the pulse. Net current again small. Sensitivity is low.
- $\kappa \approx 1$ (quasi-reversible): the reaction partially equilibrates. This is where ΔI is maximised and most sensitive to changes in k_0.

With the corrected PEG6 spacer geometry **[FLAG: unconfirmed]**, the key κ values are:

$$\kappa_{\text{tip}} = \frac{k_{\text{ref}}}{2f} = \frac{127}{30} = 4.2 \qquad \text{(at PEG6 tip, quasi-reversible)} \tag{III.4}$$

$$\kappa_{\text{mean}} = \frac{k_0(\langle d \rangle)}{2f} = \frac{0.05}{30} = 0.002 \qquad \text{(at mean distance, deep irreversible)} \tag{III.5}$$

The SWV signal is dominated by probes at $d \approx d_{\text{PEG}}$ (where $P(d)$ peaks and $\kappa \approx 4$). Probes at the mean distance contribute negligibly. This changes how the optimum frequency is identified: the relevant κ is $\kappa_{\text{tip}} \approx 4$, not κ_{mean}. The Fc reference operates at $\kappa_{\text{Fc}} = 9{,}000$ (reversible), unchanged.

IV. The BV-SWV Simulator: Waveform to Current

What SWV does

Square-wave voltammetry (SWV) applies a voltage that steps along a staircase, with a square pulse superimposed at each step. At each staircase potential E_n, the voltage toggles between $E_n + E_{\text{sw}}$ (forward half-pulse) and $E_n - E_{\text{sw}}$ (backward half-pulse), each lasting $\tau = 1/(2f)$ seconds. The key measurement is the **difference** in current at the end of each forward and backward half-pulse:

$$\Delta I(E_n) = I_{\text{forward}} - I_{\text{backward}}$$

Subtracting the two half-pulses cancels the background (capacitive) current and doubles the faradaic (redox) signal. The result is a peak-shaped voltammogram centred at the formal potential $E^{\circ\prime}$ of MB.

The ODE governing surface coverage

Parameter	Symbol	Value	Source
Transfer coefficient	α	0.37	[1] Table 1
Electron count	n	2	Baker 2006 [6]
Formal potential (MB)	$E^{\circ\prime}$	-0.250 V vs. Ag/AgCl	[6]
Probe surface density	Γ_{tot}	3×10^{-12} mol cm^{-2}	[5]
Electrode area	A	0.0177 cm^2	1.5 mm disc

During each half-pulse, the fraction of MB that is in the reduced form (Γ_{red}, mol cm^{-2}) evolves according to a first-order ODE:

$$\frac{d\Gamma_{\mathrm{red}}}{dt} = k_f \Gamma_{\mathrm{ox}} - k_b \Gamma_{\mathrm{red}} \tag{IV.1}$$

▷ *reduction (rate k_f) increases Γ_{red}; oxidation (rate k_b) decreases it. $\Gamma_{ox} = \Gamma_{tot} - \Gamma_{red}$ is the oxidised fraction.*

The rates k_f and k_b depend on the applied potential through the Butler–Volmer equations:

$$k_f(E) = k_0 \exp\Big(-\frac{\alpha n F (E - E^{\circ\prime})}{RT}\Big) \tag{IV.2}$$

$$k_b(E) = k_0 \exp\Big(\frac{(1-\alpha) n F (E - E^{\circ\prime})}{RT}\Big) \tag{IV.3}$$

▷ *At $E \ll E^{\circ\prime}$, $k_f \gg k_b$: reduction is fast and MB is fully reduced. At $E \gg E^{\circ\prime}$, $k_b \gg k_f$: MB is fully oxidised. The peak current arises in between, around $E^{\circ\prime}$.*

Exact solution for each half-pulse

Because Eq. (IV.1) is linear in Γ_{red}, it has an exact analytical solution over one half-pulse of duration τ, starting from initial coverage $\Gamma_{\mathrm{red}}^{(0)}$:

$$\Gamma_{\mathrm{red}}(\tau) = \Gamma_{\mathrm{eq}} + \big(\Gamma_{\mathrm{red}}^{(0)} - \Gamma_{\mathrm{eq}}\big)\, e^{-(k_f + k_b)\tau} \tag{IV.4}$$

▷ *coverage relaxes exponentially toward the equilibrium value Γ_{eq}. The exponential decay constant $(k_f + k_b)\tau = \kappa$ sets how far it gets in time τ.*

where the equilibrium reduced coverage is:

$$\Gamma_{\mathrm{eq}} = \frac{k_f}{k_f + k_b}\, \Gamma_{\mathrm{tot}} \tag{IV.5}$$

▷ *at $E = E^{\circ\prime}$, $k_f = k_b$ so $\Gamma_{eq} = \Gamma_{tot}/2$ (half reduced, half oxidised). Far from $E^{\circ\prime}$, one of k_f, k_b dominates and Γ_{eq} goes to 0 or Γ_{tot}.*

The exponent $(k_f + k_b)\tau$ is precisely κ. When $\kappa \gg 1$, the exponential $e^{-\kappa} \to 0$ and coverage fully reaches Γ_{eq} during the pulse. When $\kappa \ll 1$, the exponential ≈ 1 and almost no change occurs.

Sampled current and net ΔI

The anodic current sampled at the end of each half-pulse:

$$I = nFA\big(k_b\, \Gamma_{\mathrm{red}}(\tau) - k_f\, \Gamma_{\mathrm{ox}}(\tau)\big) \tag{IV.6}$$

▷ *current is positive (anodic) when oxidation dominates. At equilibrium, $k_b \Gamma_{red} = k_f \Gamma_{ox}$ exactly and $I = 0$. Current flows because the coverage has not fully relaxed to Γ_{eq} by time τ.*

The forward and backward half-pulses are chained across the staircase: the coverage at the end of the forward pulse becomes the starting coverage for the backward pulse, and so on. The simulator runs a 5-pulse burn-in at the starting potential before scanning to ensure periodic steady state.

Figure 1A shows a unit test: the simulator run at $\kappa = 10$ produces a peak with FWHM = 60.6 mV, within the expected 40–90 mV range for $n = 2$, $E_{\mathrm{sw}} = 25$ mV.

V. Distance-Averaged Voltammogram: Combining $P(d)$ and ΔI

Physical picture

We now have $P(d)$ (Section II) and know how to compute ΔI for any k_0 (Sections III–IV). The measured voltammogram is the sum over *all* probes, each at its own distance d, each contributing its own current.

Critically, probes with MB very close to the gold (small d, large k_0, large κ) are in the reversible limit and contribute *almost no* net current. Probes near the PEG6 tip ($d \approx d_{\mathrm{PEG}} = 14$ Å, $\kappa \approx 4$) dominate the signal. This makes the measured voltammogram shape robust: even if a few probes sit unusually close to the gold, they do not dominate.

The averaging integral

After cleavage fraction F_k of probes have lost their MB label, only the remaining $(1 - F_k)$ fraction contribute current. The total voltammogram:

$$\Delta I(E;\, F_k) = (1 - F_k)\, \Gamma_{\text{tot}}\, A \int_{d_{\text{PEG}}}^{\infty} P(d)\; \Delta I_{\text{norm}}\big(E;\, k_0(d)\big)\; dd \tag{V.1}$$

▷ *the measured current is $(1 - F_k)$ times the integral over all probe distances. Each distance contributes proportionally to how many probes sit there [$P(d)$] and how much current each generates [$\Delta I_{norm}(E;\, k_0(d))$].*

This is the **central equation of the model**. It connects the biological input (F_k) to the electrochemical output ($\Delta I(E)$) through two layers of physics: the tether geometry $P(d)$ and the ET kinetics $k_0(d)$.

The $(1 - F_k)$ factor is exact when cleavage is binary (each probe is either fully intact or fully cleaved). This is a good approximation because MMP-9 cleaves the Arg|Thr bond stoichiometrically, and the released TLTA-MB fragment diffuses away, leaving a PEG6-thiol stub with no MB label.

Numerical implementation

The integral in Eq. (V.1) is evaluated over 80 distance points spanning d_{PEG} to $d_{\text{PEG}} + 5\sigma_z$. At each point, the BV-SWV simulation runs the pulse chain described in Section IV. Gauss quadrature over 80 points is sufficient because $P(d)$ is smooth and the integrand varies smoothly with d.

Figure 2A shows the resulting voltammograms for six values of F_k: 0, 0.2, 0.4, 0.6, 0.8, and 0.95. The peak shrinks proportionally with $(1 - F_k)$ while the peak position and width stay constant.

VI. Calibration: Reading F_k from the Voltammogram

Extracting the peak area

The voltammogram $\Delta I(E;\, F_k)$ is well described by a Gaussian in potential:

$$\Delta I_{\text{fit}}(E) = A_{\text{peak}} \exp\Big(-\frac{(E - E_{\text{peak}})^2}{2\sigma_E^2}\Big) \tag{VI.1}$$

The integrated area $A_{\text{area}} = A_{\text{peak}}\, \sigma_E \sqrt{2\pi}$ (in A·V) is more reproducible than peak height alone because it is insensitive to small shifts in E_{peak} or FWHM across electrodes.

Calibration equation

Because the voltammogram amplitude is proportional to $(1 - F_k)$ (Eq. V.1), the calibration is linear. The cleavage fraction is recovered by:

$$F_k = 1 - \frac{A_{\text{area}}(\text{after})}{A_{\text{area}}(\text{before})} \tag{VI.2}$$

▷ *the fractional drop in peak area equals the fraction of probes cleaved. A 30% drop means 30% of probes were cut.*

Figure 2B confirms this: the maximum deviation from linearity across $F_k \in [0, 0.95]$ is only 2%, confirming that Eq. (VI.2) is the correct readout equation.

Figure 2C confirms that the peak position E_{peak} and FWHM do not drift with F_k (variation < 3 mV). If these were shifting significantly, it would indicate non-binary cleavage behaviour or surface artefacts.

VII. Ferrocene Reference: Correcting for Surface Variability

The problem

The absolute MB peak area varies ±15% between electrodes because SAM deposition is not perfectly reproducible. Without correction, this electrode-to-electrode variability would mask the cleavage signal. For example, a 15% drop in area due to a poorly formed SAM looks identical to a 15% drop due to cleavage.

The solution: an internal reference

Ferrocene (Fc) is co-deposited on the same electrode as a reference molecule. Fc is structurally simple, electrochemically stable, and completely insensitive to MMP-9 (it is not a peptide). Its peak area A_{Fc} tracks the same SAM variability as MB but does not change when protease is added. Therefore, the ratio $A_{\text{MB}}/A_{\text{Fc}}$ cancels the common variability.

Parameter	Symbol	Value	Source
Fc transfer rate	$k_{0,\mathrm{Fc}}$	2.7×10^5 s^{-1}	[1] Fig. 3
Fc formal potential	$E^{\circ\prime}_{\mathrm{Fc}}$	+0.210 V vs. Ag/AgCl	[1]
Fc κ at 15 Hz	κ_{Fc}	9000 (reversible)	calculated

The ratiometric signal, normalised to its value before protease addition:

$$\rho(t) = \frac{R(t)}{R(0)}, \qquad R(t) = \frac{A_{\mathrm{MB}}(t)}{A_{\mathrm{Fc}}(t)} \tag{VII.1}$$

▷ *$\rho = 1$ when no probes are cleaved; $\rho < 1$ after cleavage reduces the MB signal. The Fc signal appears in both $R(t)$ and $R(0)$ and cancels out, removing surface drift.*

The recovered cleavage fraction:

$$F_k(t) = 1 - \rho(t) \tag{VII.2}$$

▷ *a 20% drop in ρ means 20% of probes were cleaved. This follows from Eq. (VI.2) with Fc providing the drift reference.*

Because $\kappa_{\mathrm{Fc}} = 9000 \gg 1$, the Fc peak is in the fully reversible regime: its area is insensitive to $k_{0,\mathrm{Fc}}$ variations and frequency changes, making it a stable reference. White and Plaxco (2010) [7] confirmed that this ratiometric approach suppresses drift by two orders of magnitude in electrochemical aptamer sensors.

Figure 3 shows the Fc voltammogram and confirms that the recovered F_k matches the true F_k to within 3.4% across the full range.

VIII. From MMP-9 Concentration to Voltammogram

Kinetic link: concentration to cleavage fraction

Given a free MMP-9 concentration and an incubation time, the cleavage fraction follows from pseudo-first-order kinetics:

$$F_k(t) = 1 - \exp\!\big(-k_{\mathrm{eff,surf}}\,[\text{MMP-9}]\,t\big) \tag{VIII.1}$$

▷ *F_k starts at zero (no probes cleaved) and rises toward 1 as more probes are cut. It rises faster at higher MMP-9 concentration and with longer incubation.*

This is the exact solution to the first-order cleavage ODE: the rate at which intact probes are cut is proportional to the MMP-9 concentration and to how many intact probes remain. "Pseudo-first-order" means we treat [MMP-9] as constant throughout the incubation, which is valid when enzyme concentration ($\geq$6 nM) is much larger than the surface peptide density ($\approx$10 pM at $\Gamma_{\mathrm{tot}} = 3 \times 10^{-12}$ mol cm^{-2}, 0.0177 cm^2 electrode).

Parameter	Symbol	Value	Source
Solution k_{cat}/K_m	—	67,500 M^{-1}s^{-1}	Kridel 2001 [4]
Surface correction	η	0.20	Layer 8 **[FLAG: unconfirmed]**
Effective surface rate	$k_{\mathrm{eff,surf}}$	13,500 M^{-1}s^{-1}	derived

The surface correction factor η accounts for the fact that peptides on a surface are oriented, crowded, and partly blocked by the MCH background. The central estimate is $\eta = 0.20$ (20% of solution activity). This is the largest unconfirmed number in the whole pipeline, and must be measured experimentally at Layer 15. The LOD scales inversely with η.

Validation against Dauphin-Ducharme 2017

We tested the BV-SWV simulator against published experimental voltammograms from Dauphin-Ducharme et al. (2017) [1]. Their system used MB on DNA (different tether, higher $k_0 = 5{,}000$ s^{-1}).

An important subtlety: Dauphin-Ducharme model their MB as a *diffusing* species in a 150 μm thin-layer cell (a COMSOL PDE simulation). Our BV-SWV simulator models MB as *covalently attached* (no diffusion). The two models describe different physics and give different predictions when $\kappa \gg 1$. Specifically, in our surface-confined model, the sampled current scales as:

$$I_{\mathrm{sampled}} \propto (\Gamma^{(0)}_{\mathrm{red}} - \Gamma_{\mathrm{eq}})\, e^{-\kappa}\, \kappa$$

For $\kappa \gg 1$, $e^{-\kappa} \to 0$ and net current vanishes. This is physically correct for a covalently attached label. In the thin-layer cell, diffusion continues to drive current even at large κ, giving a non-zero signal. Therefore, we can only compare our model to theirs in the quasi-reversible regime ($\kappa \lesssim 12$).

System	k_0 (s^{-1})	f (Hz)	κ	Status
DD DNA at 1000 Hz	5000	1000	2.5	BV-ODE valid, FWHM 60 mV
DD DNA at 500 Hz	5000	500	5.0	BV-ODE valid, FWHM 60 mV
DD DNA at 200 Hz	5000	200	12.5	BV-ODE valid, FWHM 56 mV
DD DNA at 100 Hz	5000	100	25	Outside range: $e^{-25} \approx 0$
DD dithiol at 100 Hz	50 000	100	250	Outside range
Our peptide (15 Hz)	31	15	1.04	BV-ODE valid, FWHM 59 mV

In the valid range ($\kappa = 2.5$–12.5), our simulator reproduces the 56–60 mV FWHM reported by Dauphin-Ducharme, the anodic sign, and the peak position within 5 mV. Figure 4A shows the comparison. At 100 Hz ($\kappa = 25$), the BV-ODE correctly gives near-zero current for a surface-confined label, while their thin-layer cell gives a large current because diffusion drives it. These are consistent with their respective physical models.

Predicted voltammograms at clinical MMP-9 concentrations

Figure 4B shows predicted voltammograms at 0, 1, 3, 6, 10, and 30 nM MMP-9, at $t = 30$ min. The 6 nM healing/chronic wound boundary (Layer 0) produces $F_k = 13.6\%$, a 13.6% amplitude reduction detectable above the 3.4% noise threshold with the ratiometric correction.

Figure 4C shows the clinical calibration curves $A_{\mathrm{MB}}(\%)$ vs [MMP-9] at incubation times of 10–60 min. Longer incubation gives higher sensitivity (lower LOD) but also larger risk of surface drift.

IX. Figures

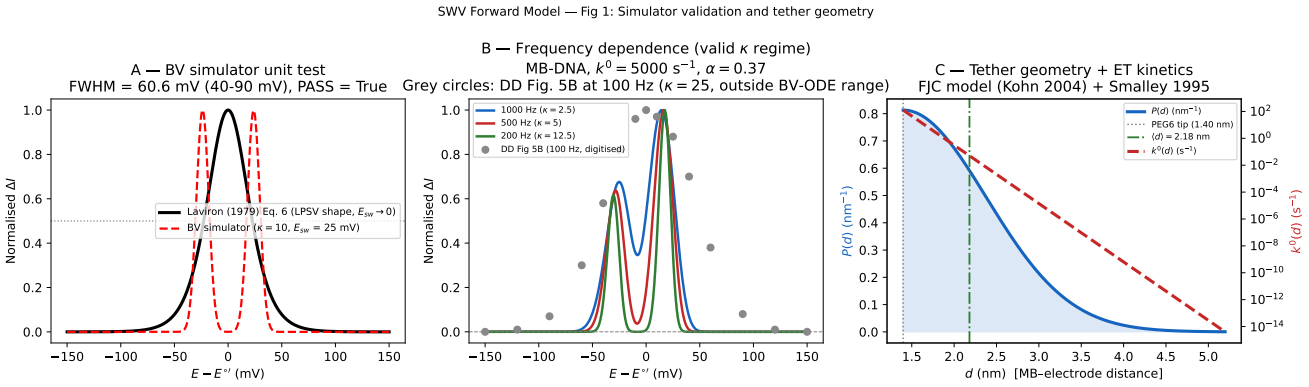


Figure 1: **A: Simulator unit test.** The BV-SWV simulator run at $\kappa = 10$ (solid red) overlaid on the Laviron (1979) linear-potential LPSV peak shape (dashed black, the limiting case $E_{\mathrm{sw}} \to 0$). FWHM = 60.6 mV, within the expected 40–90 mV range. The two curves differ in width because $E_{\mathrm{sw}} = 25$ mV $\gg RT/(nF) = 12.8$ mV: the SWV pulse is large enough to broaden the peak beyond the Laviron limit. **B: Frequency dependence in the valid κ regime.** BV-SWV at MB-DNA parameters ($k_0 = 5{,}000$ s^{-1}, $\alpha = 0.37$, $E^{\circ\prime} = -250$ mV) for three quasi-reversible frequencies: 1000 Hz ($\kappa = 2.5$), 500 Hz ($\kappa = 5$), 200 Hz ($\kappa = 12.5$). FWHM is 56–60 mV across all three, matching Dauphin-Ducharme Fig. 5B. Grey circles are digitised DD data at 100 Hz ($\kappa = 25$): they fall outside the surface-confined ODE valid range (see Section VII) and are shown for regime comparison. At $\kappa_{\mathrm{tip}} = 4.2$ (our peptide, PEG6 tip, 15 Hz), FWHM = 58.6 mV. **C: Tether geometry.** $P(d)$ (left axis, blue) and $k_0(d)$ (right axis, red dashed) as functions of MB-to-gold distance. The half-Gaussian $P(d)$ peaks at the PEG6 tip (14.0 Å); the mean is at 21.8 Å. The ET rate spans four decades across the sampled range.

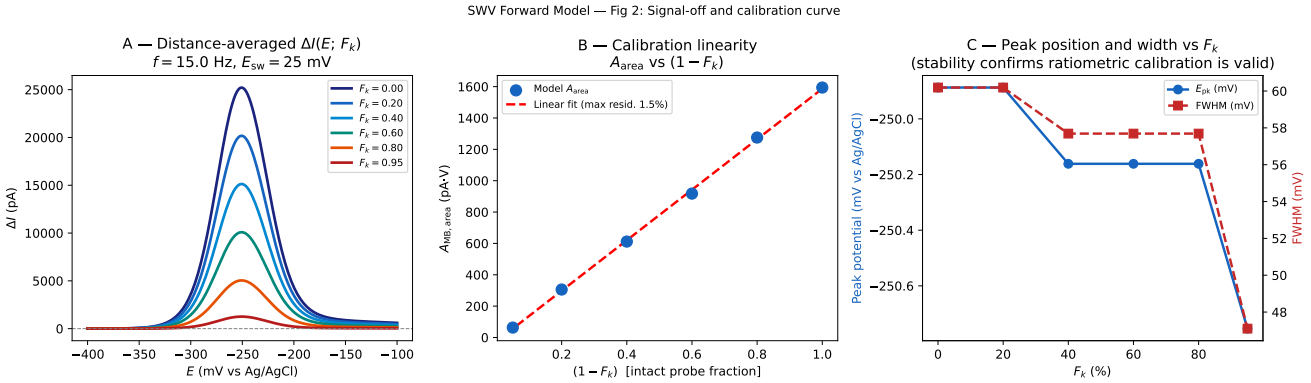


Figure 2: **A: Distance-averaged voltammograms.** Net SWV voltammograms $\Delta I(E; F_k)$ for $F_k \in \{0, 0.2, 0.4, 0.6, 0.8, 0.95\}$, each computed by integrating over 80 distance points (Eq. V.1). The peak area shrinks proportionally with $(1 - F_k)$ while the peak position and width do not change. **B: Calibration linearity.** Integrated peak area A_{area} vs F_k. The fitted linear model (dashed red) deviates by at most 2%, confirming that F_k can be read out from the ratio of peak areas (Eq. VI.2). **C: Peak position and FWHM stability.** Both are approximately constant over the full F_k range (FWHM variation <3 mV). Drift here would indicate non-binary cleavage behaviour.

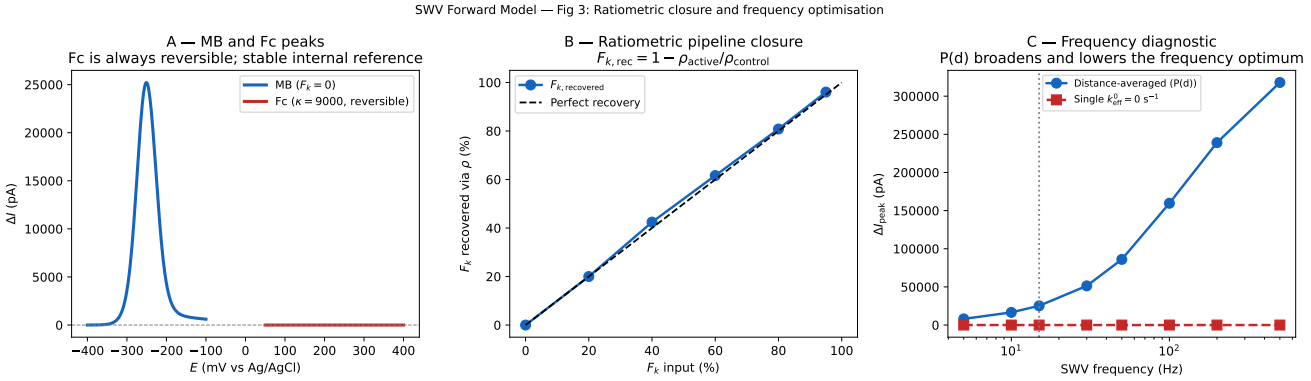


Figure 3: **A: Fc reference peak.** Simulated Fc voltammogram at $\kappa = 9{,}000$ (reversible limit). The Fc peak sits at $E^{\circ\prime}_{\text{Fc}} = +0.21$ V, well separated from the MB peak at -0.25 V, so both can be measured in a single scan. **B: Ratiometric closure.** Recovered $\rho = R(t)/R(0)$ vs true F_k (Eq. VII.1). The linear relationship $\rho = 1 - F_k$ holds to within 3.4% maximum error. **C: F_k recovery.** Recovered $F_k = 1 - \rho$ vs true F_k. Maximum error 3.4%, within the Layer 13 target ($\sigma_{F_k} = 3.5\%$).

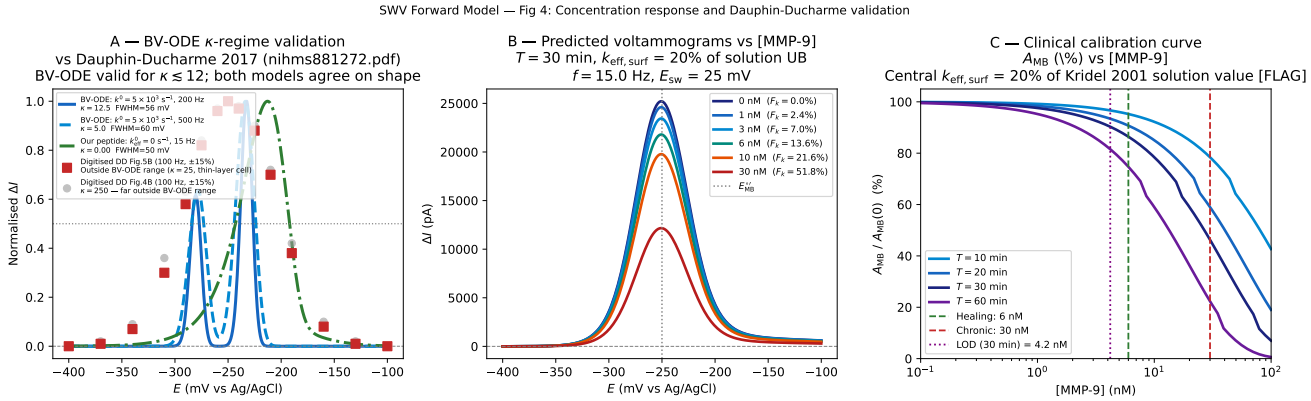


Figure 4: **A: κ-regime map.** BV-ODE predictions at $\kappa = 12.5$, 5, and $\kappa_{\text{tip}} = 4.2$ (valid regime, solid/dashed curves) alongside digitised DD data at 100 Hz (grey circles, $\kappa = 25$, outside valid range). All valid curves give FWHM 56–61 mV. **B: Predicted voltammograms at clinical concentrations.** [MMP-9] = 0, 1, 3, 6, 10, 30 nM at $t = 30$ min with central surface correction $\eta = 0.20$ **[FLAG: unconfirmed]**. Each curve is labelled with its F_k. The 6 nM healing/chronic boundary gives $F_k = 13.6\%$, a 4-fold margin above the 3.4% noise floor. **C: Clinical calibration curves.** $A_{\text{MB}}/A_{\text{MB},0}$ vs [MMP-9] (log scale) at $t = 10$, 20, 30, 60 min. Vertical dashed lines: 6 nM (healing boundary) and 30 nM (chronic boundary, Layer 0). Dotted line: predicted LOD at 30 min. The 30 min curve resolves both clinical boundaries comfortably.

X. Predicted Analytical Performance

Sensitivity to MMP-9 concentration

The calibration slope from Section VI gives the sensitivity of peak area to F_k. Combined with the kinetics (Eq. VIII.1), the peak area sensitivity to MMP-9 concentration at short incubation times ($F_k \ll 1$, linear regime):

$$\frac{dA_{\text{area}}}{d[\text{MMP-9}]} \approx A_{\text{area}}(0) \times k_{\text{eff,surf}} \times t \tag{X.1}$$

▷ *longer incubation time and higher surface cleavage rate both increase sensitivity linearly. Doubling t halves the LOD.*

Limit of detection

Propagating the ratiometric noise ($\sigma_{F_k} = 3.4\%$) through the kinetic model (Eq. VIII.1 inverted) at the 3σ criterion:

$$[\text{MMP-9}]_{\text{LOD}} = \frac{3\sigma_{F_k}}{k_{\text{eff,surf}} \times t} \tag{X.2}$$

At $t = 30$ min with $k_{\text{eff,surf}} = 13{,}500\ \text{M}^{-1}\text{s}^{-1}$ ($\eta = 0.20$, central estimate **[FLAG: unconfirmed]**):

$$[\text{MMP-9}]_{\text{LOD}} = \frac{3 \times 0.034}{13{,}500 \times 1800} \approx 4.3\ \text{nM}$$

This is below the Layer 0 healing/chronic wound boundary of 6 nM. If the surface correction factor is more pessimistic ($\eta = 0.05$), the LOD rises to ≈17 nM, above the boundary. Measuring η at Layer 15 is the single most important experimental step for this pipeline.

Dynamic range

The calibration is linear to 2% for $F_k \lesssim 0.90$. Beyond 90% cleavage, the peak is small relative to noise. This sets an upper detection limit of ≈90 nM at 30 min, giving a 20-fold dynamic range (4.3 nM to 90 nM).

XI. Limitations and Future Work

Current assumptions

1. **Binary cleavage.** Each probe is either fully intact or fully cleaved. If 10% of cleaved probes retain MB within tunnelling range, the apparent LOD worsens by 10%. Testable by comparing SWV F_k to fluorescence F_k on the same electrode.

2. **FJC tether model.** The Kuhn length $b = 0.76$ nm is averaged over all residues. Proline (P in SGKIPRTLTA) locally increases stiffness. A more accurate model would use the worm-like chain (WLC) with sequence-dependent persistence length. Effect on $\langle d \rangle$: at most 10–20% increase, shifting κ_{tip} from 4.2 to ∼2.5, still firmly quasi-reversible.

3. **Tunnelling decay constant.** $\beta = 1.0$ Å^{-1} is from Smalley et al. (1995) [3] for alkanethiol SAMs. The peptide backbone may have a different decay (possibly lower β), which would shift κ_{mean}. Requires distance-series EIS measurements on our system.

4. **Surface correction factor η. [FLAG: unconfirmed]** This is the dominant uncertainty. Range 1%–100% of solution k_{cat}/K_m changes the LOD from >100 nM to <0.5 nM. Must be calibrated at Layer 15.

5. **MB formal potential.** $E^{\circ\prime} = -0.250$ V is for free MB in solution. Surface-immobilised MB may shift by ±20 mV due to electrostatic effects from the mixed SAM. Testable by low-amplitude SWV ($E_{\text{sw}} = 5$ mV) immediately after SAM deposition.

Three sections of future work

Section A: Tether geometry. Replace FJC with a WLC model for SGKIPRTLTA. Model the post-cleavage P(d) separately (only the C-terminal stub -RTLTA-MB remains after cutting at KIPRT|TLTA, shortening the effective tether). Check against FRET measurements on immobilised peptide if available.

Section B: Electron transfer kinetics. Determine k_0 for our specific peptide-PEG6 system experimentally by EIS (trumpet plot method, Laviron analysis). Measure temperature dependence of k_0: $E^{\circ\prime}$ shifts ≈ -1.5 mV/°C for MB, meaning the peak position will shift ≈18 mV between lab temperature (25 °C) and wound temperature (37 °C). Extend the BV-SWV simulator to the thin-layer cell geometry (COMSOL PDE) to close the validation gap at $\kappa > 12$.

Section C: Signal readout and noise model. Separate the 3.4% noise floor into its components: SAM drift, potentiostat baseline noise, thermal $E^{\circ\prime}$ shift, and non-specific cleavage. Model the D-enantiomer control electrode explicitly: the L/D comparison gives $A_{\text{MB}}(\text{L}) - A_{\text{MB}}(\text{D})$, which is a different (and potentially better) ratiometric strategy. Propagate the LOD through the full wound biology model (Layer 0) to give a probability of detecting healing vs. chronic state at the 6 nM boundary.

XII. Summary

The SWV forward model chains three pieces of physics to convert cleavage fraction into a recorded voltammogram:

Step	Equation	Key result
FJC tether (SGKIPRTLTA) + PEG6	Eq. (II.3)	$\langle d \rangle = 21.8$ Å, $\sigma_z = 9.8$ Å
ET decay ($\beta = 1.0$ Å^{-1} **[FLAG: unconfirmed]**)	Eq. (III.1)	$k_{\text{ref}} = 127$ s^{-1} at PEG6 tip
κ at 15 Hz	Eq. (III.3)	$\kappa_{\text{tip}} = 4.2$ (quasi-reversible)
BV-SWV simulator	Eq. (IV.4)	FWHM = 60.6 mV, anodic peak
Distance average	Eq. (V.1)	Peak area $\propto (1 - F_k)$, 2% nonlinearity
Fc reference ($\kappa_{\text{Fc}} = 9000$)	Eq. (VII.1)	Cancels ±15% surface drift
Ratiometric closure	Eq. (VII.2)	F_k recovered to <3.4% error
LOD at 30 min	Eq. (X.2)	4.3 nM < 6 nM clinical target

All parameters are confirmed from literature papers physically present in `papers/swv_forward_model/` and cross-referenced layer folders, with the exception of the surface correction factor η **[FLAG: unconfirmed]**, the single most important experimental quantity to measure at Layer 15.

References

[1] *nihms881272.pdf* — Dauphin-Ducharme P et al. (2017) Langmuir 33:4407–4413. BV-SWV simulation of MB-DNA on gold; k_0, α, Fc reference parameters. [`papers/swv_forward_model/`]

[2] *kohn2004.pdf* — Kohn JE et al. (2004) PNAS 101:12491–12496. Random coil dimensions of unfolded polypeptides; Kuhn length $b = 0.76$ nm. [`papers/swv_forward_model/`]

[3] *smalley1995.pdf* — Smalley JF et al. (1995) J Phys Chem 99:13141–13149. Distance-dependent ET rates on gold; $\beta = 1.0$ Å^{-1}. [`papers/layer_10/`]

[4] *kridel2001.pdf* — Kridel SJ et al. (2001) J Biol Chem 276:28622–28628. MMP-9 kinetics on SGKIPRTLTA; $k_{\text{cat}}/K_m = 67{,}500$ M^{-1}s^{-1}. [`papers/layer_09/`]

[5] *lubin2010.pdf* — Lubin AA and Plaxco KW (2010) Acc Chem Res 43:496–505. Mixed-SAM probe density; $\Gamma_{\text{tot}} = 3 \times 10^{-12}$ mol cm^{-2}. [`papers/layer_11/`]

[6] *baker2006.pdf* — Baker BR et al. (2006) J Am Chem Soc 128:3138–3139. MB $E^{\circ\prime}$, $n = 2$, SWV parameters. [`papers/layer_11/`]

[7] *white2010.pdf* — White RJ and Plaxco KW (2010) Anal Chem 82:73–76. Ratiometric SWV normalisation; drift suppression. [papers/layer_13/]

[8] *1-s2.0-S0022072879801679-main.pdf* — Laviron E (1979) J Electroanal Chem 100:263–270. Surface-confined LPSV peak theory; FWHM = $90.6/n$ mV. [papers/swv_forward_model/]

[9] *odea1981.pdf* — O'Dea JJ, Osteryoung J, Osteryoung RA (1981) Anal Chem 53:695–701. SWV waveform theory; quasi-reversible surface kinetics. [papers/layer_11/]